

Expresar en PF

i) $+0x8452 \Rightarrow +1 \overset{\text{Asar a Bin}}{000} 0100 0101 0010 * 2^0 \Rightarrow$
 $1,000 0100 0101 0010 * 2^{15}$

$EXP = 15 + \text{sesgo} = 15 + 127 = 142 \Rightarrow 1000 1110$

+ Exp+sesgo MANTISA sm 1 inicial

0	1000 1110	000 0100 0101 0010	0000 0000
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ii) $133 \Rightarrow +1000 0101 * 2^0 \Rightarrow 1,0000 101 * 2^7$

$EXP = 7 + 127 = 134 \Rightarrow 1000 0110$

+ Exp+sesgo MANTISA sm 1 inicial

0	1000 0110	000 0101 0000 0000	0000 0000
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Expresar en decimal

i) $0x8B50-0000 \Rightarrow \overset{\text{Sig}}{1} \overset{\text{EXP+SESgo}}{011 1011} \overset{\text{MANT}}{0101 0000 0000 0000 0000 0000}$
 $\ominus$

$EXP = 01110110 = 118 = \text{sesgo} = 118 - 127 = -9$

$MANTISA = (1,101)_2 = 1 \cdot 2^0 + 1 \cdot 2^{-1} + 0 \cdot 2^{-2} + 1 \cdot 2^{-3} = 1 + 0,5 + 0,125 = 1,625$ $\left\{ \ominus 1,625 \cdot 2^{-9} \right\}$

ii) $0x8040-0000 \Rightarrow \overset{\text{Sig}}{1} \overset{\text{EXP+SESgo}}{0000 0000} \overset{\text{MANT}}{0100 0000 0000 0000 0000 0000}$ Número denormalizado

$EXP = 00000000 = 1 - 127 = -126$
 $MANTISA = 2^{-1} = 0,5$ $\left\{ \ominus 0,5 \cdot 2^{-126} \right\}$

iii) $0x80C0-0000 \Rightarrow \overset{\text{Sig}}{1} \overset{\text{EXP+SESgo}}{0000 0000} \overset{\text{MANT}}{1100 0000 0000 0000 0000 0000}$

$EXP = 00000001 = 1 - 127 = -126$
 $MANTISA = 1 + 2^{-1} = 1,5$ $\left\{ -1,5 \cdot 2^{-126} \right\}$

iv) $0xFF80-0000 \Rightarrow \overset{\text{Sig}}{1} \overset{\text{EXP+SESgo}}{1111 1111} \overset{\text{MANT}}{1000 0000 0000 0000 0000 0000}$
 $\ominus$
 $= -\infty$

v) $0x40F0-0000 \Rightarrow \overset{\text{Sig}}{0} \overset{\text{EXP+SESgo}}{1000 0000} \overset{\text{MANT}}{1111 0000 0000 0000 0000 0000}$
 $\oplus$

$EXP = 129 - 127 = 2$
 $MANTISA = 1,111 = 1,875$ $\left\{ +1,875 \cdot 2^2 \right\}$

1	8	23
S	EXP	MANTISA
	$\downarrow$	$\downarrow$
	sesgo	1. MANTISA
	$2^{n-1} - 1$	
	(127)	

$\boxed{0} \left\{ \begin{array}{l} EXP 00000000 \\ MANT 00000000 \dots \end{array} \right.$
 $\boxed{+00} \left\{ \begin{array}{l} EXP 11111111 \\ MANT 000 \dots +0 \text{ a } 111 \dots \end{array} \right.$
 $\boxed{-00} \left\{ \begin{array}{l} EXP 11111111 \\ MANT 000 \dots +0 \text{ a } 111 \dots \end{array} \right.$

EXP que empiezan por 0
 $\ominus$
 $\oplus$
 $\oplus$

$\frac{1}{10000000} \frac{1111 \dots 11}{1111 \dots 11} \cdot 2^{-126} =$
 $0,111 \dots 11 \cdot 2^{-126} =$
 $= (1 - 2^{-23}) \cdot 2^{-126} =$
 110

111111111111...11 NaN

